

Data Flow Diagram

Date	29 September 2026
Team ID	SWTID-2026-1156
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system – between external entities, processes, and data stores. Use the symbol legend below as a guide.

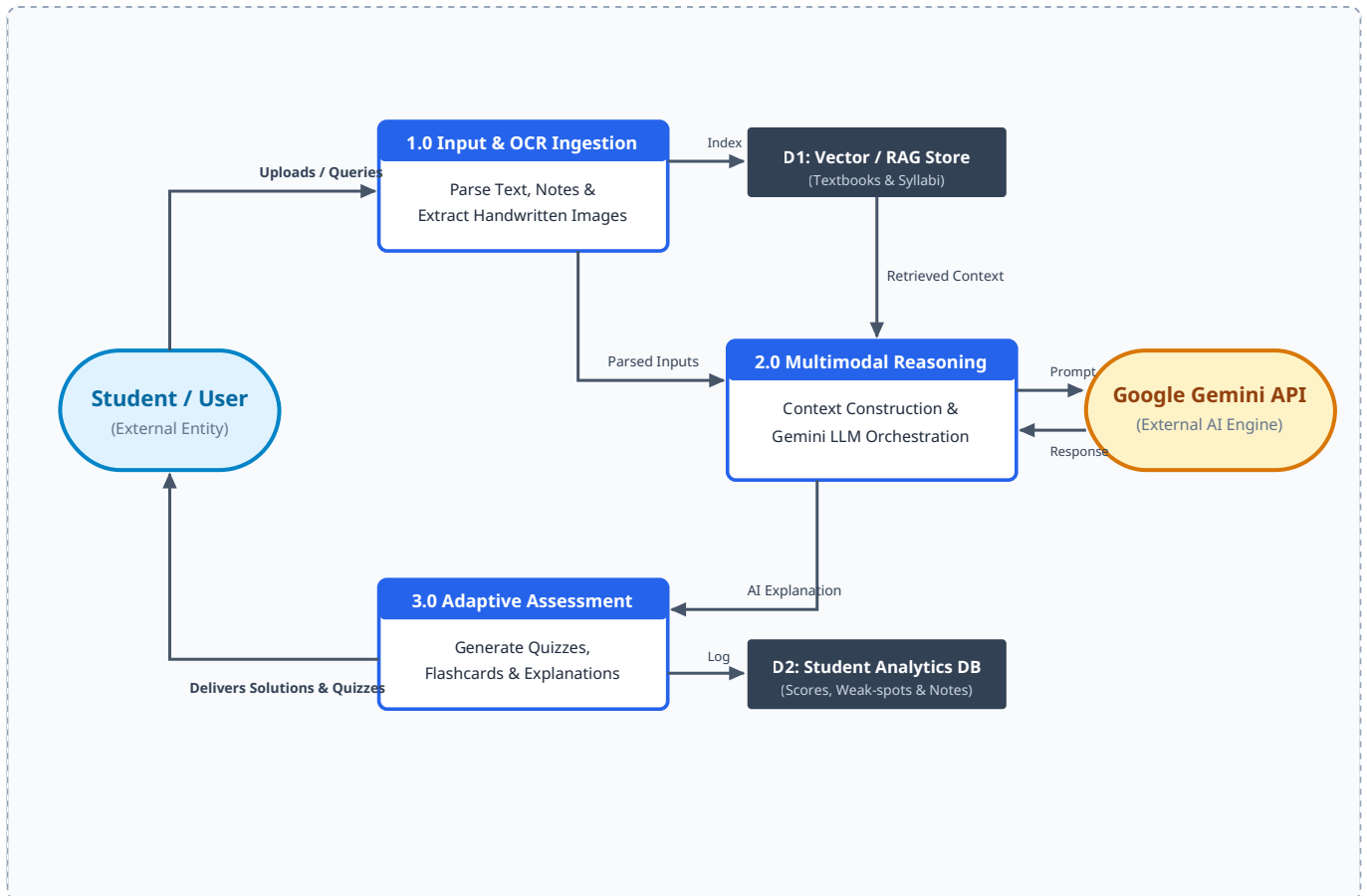
DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., <i>Student, Google Gemini API, LMS</i>).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., <i>1.0 Ingest Document & Query, 2.0 Multimodal Reasoning Engine, 3.0 Generate Diagnostic Assessment</i>).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., <i>Vector Embeddings DB, User Query History, Assessment & Flashcard Store</i>).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Example

The diagram on Page 2 illustrates how these symbols connect into an architectural Level-1 Data Flow Diagram for **EduGenie**: depicting input multimodal processing from the student, prompt enrichment with textbook embeddings, orchestration via Google Gemini API, and storage into personalized progress records.

Level-1 Data Flow Diagram (DFD) — EduGenie Architecture



Data Flow Summary: The student sends multimodal queries (handwritten notes, textbook PDFs, voice). Process **1.0** extracts and structures visual/textual data. Process **2.0** enriches prompts with syllabus context from **D1** and communicates bidirectionally with the **Google Gemini API**. Process **3.0** delivers step-by-step guidance, evaluates quizzes, and logs analytics into **D2** for continuous personalized learning.